ABHISHEK PANDA

Technical Architect | Azure, .NET & Enterprise AI Engineering

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abhishekpanda

EXECUTIVE SUMMARY

- **Technical Architect / Lead Engineer** with ~**12+** years of experience architecting and delivering scalable, cloud-based enterprise platforms using **C#, .NET Core/.NET Framework, Azure**, and **microservices** architecture across **HealthCare & Insurance, banking**, **aviation**, **e-Commerce**, **Automobile** and **SaaS** domains.
- Experienced in designing **AI-enabled enterprise platforms** using **Semantic Kernel, RAG** architectures, **Azure AI services**, and **LLM** integrations within **.NET & Python** ecosystems.
- Experienced in enterprise **platform modernization, API governance**, **cloud migration, observability, performance optimization, cost optimization,** and scalable backend engineering with strong focus on reliability, security, and maintainability.
- Proven leadership in driving engineering excellence, mentoring cross-functional teams, defining technical strategy, and delivering high-impact solutions aligned with business and digital transformation objectives.

PROFESSIONAL EXPERIENCE

QUALITEST, Technical Architect | Technical AI Architect (.NET Core) 03/2026 – Present | Bangalore

Project : **Hiscox** Insurance AI Transformation Platform

Responsibilities

- Architected scalable AI orchestration and enterprise knowledge platforms for insurance-domain automation workflows.
- Developed AI-enabled backend services using **.NET 10**, **Semantic Kernel, RAG** pipelines, and enterprise **LLM integrations**
- Designed MCP-based integration frameworks and secure AI tool orchestration architectures for modular enterprise AI ecosystems
- Led and mentored a team of 12 Machine Learning Engineers and Senior Software Engineers across architecture, delivery, and engineering governance
- Defined cloud-native architecture standards covering scalability, observability, security, DevSecOps, and enterprise AI platform reliability

Tech Stack: .NET 10 | Semantic Kernel | Agentic AI | MCP (Model Context Protocol) | Retrieval-Augmented Generation (RAG) | Azure OpenAI | Open Source LLMs | Qdrant Vector Database | Langfuse | Azure | AKS | Docker | Kubernetes | Microservices | Azure Service Bus | GitOps | LLMOps

Smart Drive Systems Pvt. Ltd., Engineering Lead (.NET) | Technical Architect 01/2023 – 08/2025 | Bangalore, India

Project : Automotive / IoT / Fleet Management– *GFP SmartDrive (Enterprise Platform)*

- Architected Azure-based microservices and event-driven systems using AKS, Azure Functions, Service Bus, APIM, and Azure SQL for scalable fleet operations
- Integrated IoT and Digital Twin technologies for real-time telemetry tracking, monitoring, and intelligent fleet management workflows
- Implemented AI/ML-based predictive analytics and optimization solutions using ML.NET and Azure Machine Learning for vehicle insights and operational efficiency
- Designed secure API ecosystems with OAuth2/JWT, API governance, fault-tolerant messaging, retry strategies, and resilient distributed architecture patterns
- Developed DevSecOps (GitOps) and Kubernetes-based cloud deployments across distributed fleet platforms
- Led & mentored 18+ engineers from USA, Mexico, Spain & India while improving platform performance and cloud cost efficiency

Tech Stack : C# | .NET Core | Azure AKS | Azure Functions | Azure SQL | Azure Service Bus | RabbitMQ | APIM | GraphQL | Angular | ML.NET | Azure Machine Learning | IoT | Azure Digital Twins | Docker | Kubernetes | Terraform | Azure Bicep | CI/CD | Azure DevOps | AWS Lambda | AWS EKS | ECR | OAuth2/JWT | Event-Driven Architecture

Accion Labs, Principal Software Engineer 06/2022 – 12/2022 | Bangalore, India

Domain: Conversational Support Systems / AI-ML Integration

KPMG (*Contract*) — Jun 2022 – Aug 2022

Accion Labs (*Converted to Full-Time*) — Aug 2022 – Dec 2022

- Developed .NET Core microservices and backend integration services for enterprise AI chatbot and conversational support platforms on Azure
- Built conversational AI workflows using ML.NET, Azure Cognitive Services (LUIS), and NLP-based intent/entity processing
- Designed secure REST API integrations with CRM and support systems using OAuth2/JWT authentication
- Implemented event-driven messaging patterns for scalable, fault-tolerant, and real-time customer interaction processing
- Optimized backend services for performance, scalability, and high-volume conversational workloads

Tech Stack :C# | .NET Core | ASP.NET Core Web APIs | Azure | Azure Service Bus | REST APIs | ML.NET | Azure Cognitive Services (LUIS) | NLP | OAuth2/JWT | Docker | CI/CD | Microservices

Wells Fargo, Assistant Vice President / Lead Engineer — 05/2020 – 07/2022 | Hyderabad, India

Wealth & Investment Management Technology (WIMT)

- Led the design and governance of **mission-critical brokerage platforms** in a regulated banking environment, implementing event-driven microservices using .NET Core, Kafka, and Azure Functions for trade processing, analytics, and real-time dashboards (Angular 12)
- Built secure and scalable **Azure cloud infrastructure** (AKS, Azure SQL, Service Bus, Key Vault, Entra ID) while contributing to architecture governance, security standards, and backend service reliability
- Managed **24x7 production operations**, SLA governance, and zero-downtime deployments (blue-green, canary), improving infrastructure efficiency, optimizing cloud costs, and mentoring 10+ engineers

Tech Stack: C#, .NET Core, Kafka, Azure AKS, Azure Functions, Azure SQL, Azure Service Bus, Key Vault, Entra ID, Angular 12, Docker, Kubernetes, CI/CD, Rabbit MQ, Graph QL.

Virtusa, Associate Consultant — 08/2018 – 06/2020 | Doha, Qatar

Qatar Airways – PAX Disruption Management System (GCC) |

Citi Bank – Financial Workflow Automation Platform

- Developed scalable microservices and serverless solutions using **.NET Core, Azure Functions, Service Bus**, and **APIM** replacing legacy monolithic systems
- Designed secure API ecosystems using OAuth2, JWT, API governance, and enterprise security standards for regulated aviation and banking environments
- Contributed to **Angular**-based dashboards, enterprise integrations, and cloud-native modernization initiatives

Tech Stack: C#, .NET Core, Azure API Management (APIM), Azure Functions, Azure Service Bus, SQL Server, OAuth2, JWT, Angular 8, REST APIs, Microservices, Event-Driven Architecture.

JD Sports Fashion LLP, Software Engineer — 04/2018 – 06/2018 | Hyderabad

Products: **JD Sports Fashion – Retail Platform Modernization**

- Developed and enhanced **ASP.NET** **Web API services** for retail operations including order processing, inventory management, and customer workflows, integrating Angular-based frontend components
- Implemented **asynchronous messaging using RabbitMQ** and contributed to Azure-based deployments (App Services, Storage) to improve system scalability and reliability
- Supported **CI/CD pipelines and DevOps practices** using Git and build/release pipelines, ensuring smooth development, integration, and deployment processes

Tech Stack: C#, ASP.NET Web API, RabbitMQ, Microsoft Azure (App Services, Storage), Angular, SQL Server, REST APIs, DevOps (CI/CD, Git)

Conduent (formerly Xerox Business Services), Associate Software Engineer — 07/2015 – 07/2017 | Bangalore, India

METLIFE – Healthcare / Insurance Systems

Bank of America (Merrill Lynch) – Financial Workflow Automation

- Developed scalable .NET Framework and ASP.NET Web API solutions supporting policy management, claims processing, underwriting, billing, and enterprise financial workflow automation across healthcare and banking domains, with secure backend integrations using Entity Framework, ADO.NET, and OAuth-based authentication.
- Contributed to early modernization initiatives by migrating legacy components to .NET Core 1.x, developing Angular-based UI modules, and supporting Azure-hosted CI/CD deployments using TFS and Git while improving application performance, reliability, and cross-platform compatibility.

Tech Stack: C#, .NET Framework 4.x, .NET Core 1.x (early adoption), ASP.NET Web API, Entity Framework, ADO.NET, REST APIs, SQL Server.

TECHNICAL SKILLS

Backend & Frontend Architecture

- **Backend Architecture :** C#, .NET Core, ASP.NET Core, Entity Framework Core, REST APIs, gRPC, GraphQL, Microservices, Kafka, SQL Server, PostgreSQL
- **Frontend Architecture :** Angular(Angular 8–18), TypeScript, REST API Integration

DevOps & Infrastructure

Git, GitHub, Azure DevOps, GitHub Actions, GitLab CI/CD, Docker, Kubernetes (AKS), Helm, Terraform, Azure Bicep, Argo CD (GitOps), CI/CD Automation, Infrastructure as Code (IaC), LLMOps .

AI / ML & Agentic Systems

Semantic Kernel, Azure AI Foundry, Azure OpenAI, LLM Integration (Open & Closed Source), RAG Architecture, Vector Databases, AI Orchestration, Multi-Agent Workflows, Tool Calling, ML.NET, Azure Machine Learning

Architecture, Security & Practices

Cloud-Native Architecture, Event-Driven Architecture, Distributed Systems,DDD, API Governance, OAuth2/JWT, DevSecOps, Secure SDLC, Agile/Scrum, Cloud Migration,Clean Architecture .

Cloud & Platform Engineering – Microsoft Azure (Primary)

AKS, Azure Functions, Azure API Management (APIM), Azure Service Bus, Event Grid, Azure SQL, Cosmos DB, Azure AI Search, Azure Key Vault, Microsoft Entra ID, Azure Monitor, Application Insights, Azure DevOps, Azure Digital Twins

EDUCATION

B.TECH - Computer Science & Engineering, — 04/2010 – 05/2014 | Odisha

Government College Of Engineering | BPUT , Odisha